EXP 17: Implement Mobile Price Prediction using appropriate machine learning algorithm

CODING:

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# Mobile Price Prediction using KNN

# Dataset: [RAM in GB, Storage in GB, Price Category]
data = [
    [2, 32, "Low"],
    [3, 32, "Low"],
    [4, 64, "Medium"],
    [6, 64, "Medium"],
    [8, 128, "High"],
    [12, 256, "High"]
]

# New mobile details
test = [8, 128]

# Calculate Euclidean distance
distances = []

for row in data:
    distance = ((test[0] - row[0]) ** 2 +
                (test[1] - row[1]) ** 2) ** 0.5

    distances.append((distance, row[2]))
```

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# Sort distances

distances.sort()


# Select 3 nearest neighbors

k = 3

neighbors = [item[1] for item in distances[:k]]


# Find majority class

prediction = max(set(neighbors), key=neighbors.count)


print("MOBILE PRICE PREDICTION")

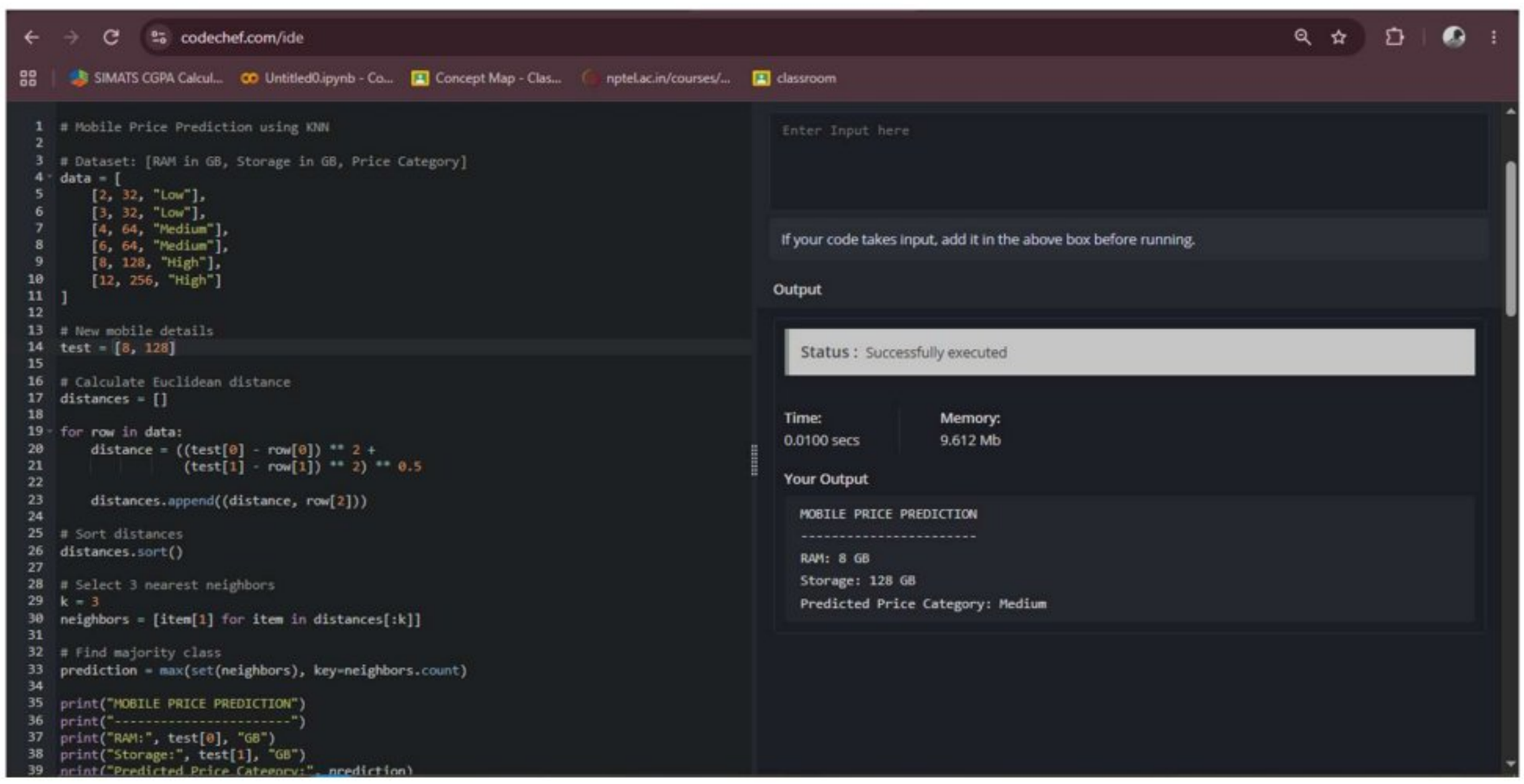
print("----------------------")

print("RAM:", test[0], "GB")

print("Storage:", test[1], "GB")

print("Predicted Price Category:", prediction)
```

OUTPUT:




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